Semester	T.E. Semester VI Div A Batch 1 – CMPN
Subject	Machine Learning Lab
Subject Professor Incharge	Prof. Kavita Shirsat
Assisting Teachers	Prof. Kavita Shirsat
Laboratory	M312A

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Roll Number	23102A0007	
Grade and Subject Teacher's Signature		

Experiment Number	01
Experiment Title	**Unsupervised and supervised learning**
Objectives (Skill Set / Knowledge Tested / Imparted)	Perform customer segmentation using K-Means clustering, determine the optimal number of clusters with the Elbow Method, assign cluster labels for analysis and supervised classification, and visualize clusters and feature relationships. Gain practical skills in unsupervised learning, K-Means, Elbow Method, data preprocessing (selection, scaling, encoding), clustering vs. classification, Random Forest training, model evaluation, visualization with matplotlib/seaborn, and awareness of data leakage.

Description	This experiment used a real-world retail dataset to perform **customer segmentation**.
	Working Principle: K-Means clustering partitions data into K clusters by minimizing the **sum of squared distances** between data points and their respective cluster centroids. The algorithm iteratively updates centroids until convergence. The **Elbow Method** is used to select the optimal K by analyzing the plot of within-cluster sum of squares.
	Procedure:
	1. Preprocess data by encoding categorical variables and selecting numerical features (Quantity, Unit Price, Total Amount). 2. Apply **K-Means clustering** to group similar transactions. 3. Determine optimal clusters using the **Elbow Method**. 4. Assign cluster labels to each data point. 5. Train a **Random Forest classifier** using demographic and product features to predict cluster labels for new data. 6. Visualize clusters and feature correlations using heatmaps and plots.

exp1ml-23102A0007.ipynb

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Upload dataset in Colab

[50]

```
from google.colab import files
uploaded = files.upload()
```

Choose Files No file chosen Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.

```
Saving retail_sales_dataset.csv to retail_sales_dataset (1).csv
```

[51]

```
import pandas as pd

df = pd.read_csv('/content/retail_sales_dataset.csv')

print(df.head())
print(df.info())
```

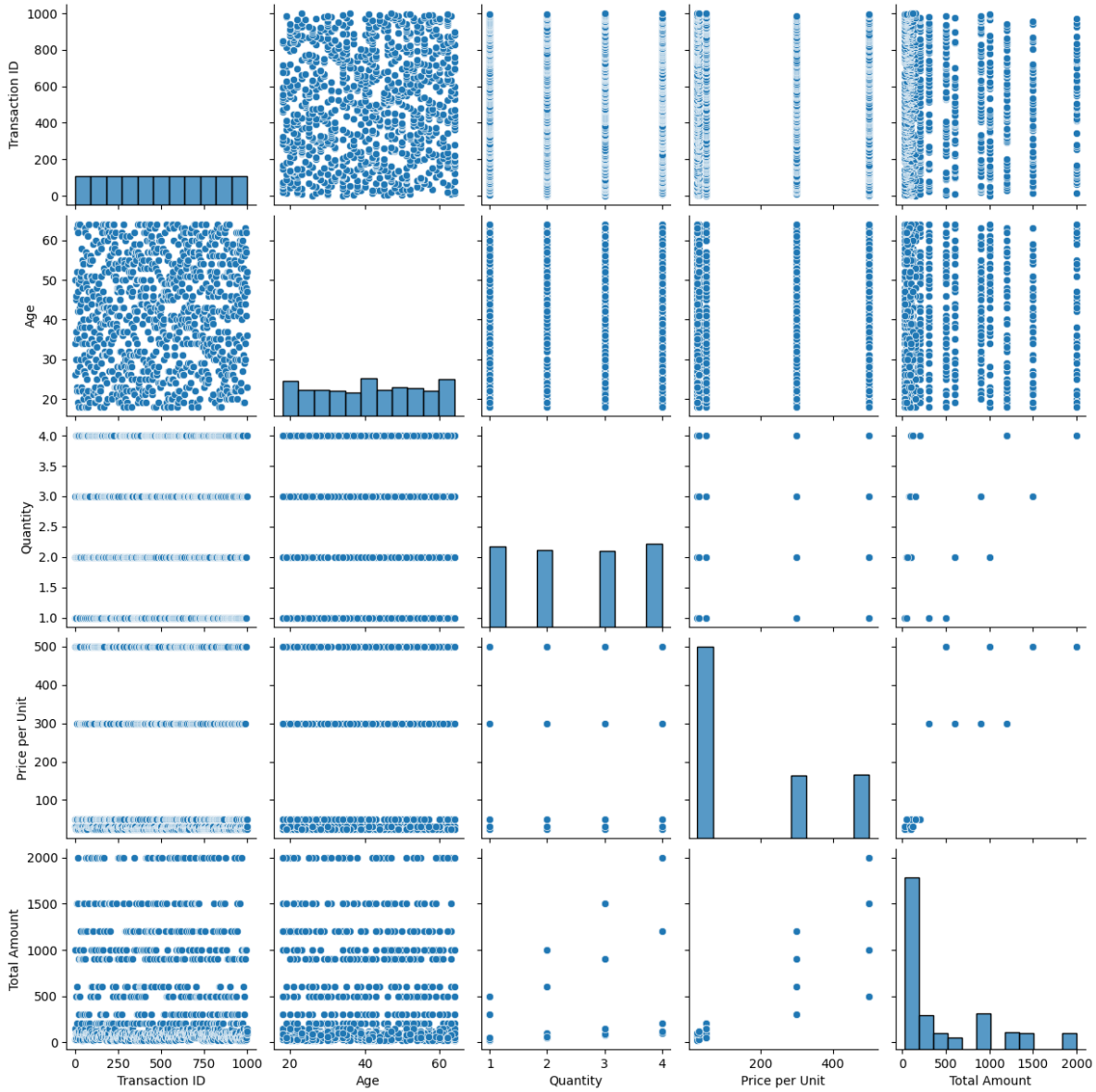
```
   Transaction ID       Date Customer ID  Gender  Age Product Category  \
0               1  2023-11-24     CUST001    Male   34           Beauty
1               2  2023-02-27     CUST002  Female   26         Clothing
2               3  2023-01-13     CUST003    Male   50      Electronics
3               4  2023-05-21     CUST004    Male   37         Clothing
4               5  2023-05-06     CUST005    Male   30           Beauty

   Quantity  Price per Unit  Total Amount
0         3              50           150
1         2             500          1000
2         1              30            30
3         1             500           500
4         2              50           100
<class 'pandas.core.frame.DataFrame'>
```

```
Data columns (total 9 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   Transaction ID    1000 non-null   int64
 1   Date              1000 non-null   object
 2   Customer ID       1000 non-null   object
 3   Gender            1000 non-null   object
 4   Age               1000 non-null   int64
 5   Product Category  1000 non-null   object
 6   Quantity          1000 non-null   int64
 7   Price per Unit    1000 non-null   int64
 8   Total Amount      1000 non-null   int64
dtypes: int64(5), object(4)
memory usage: 70.4+ KB
None
```

Pairplot (relationship between features)

[52]

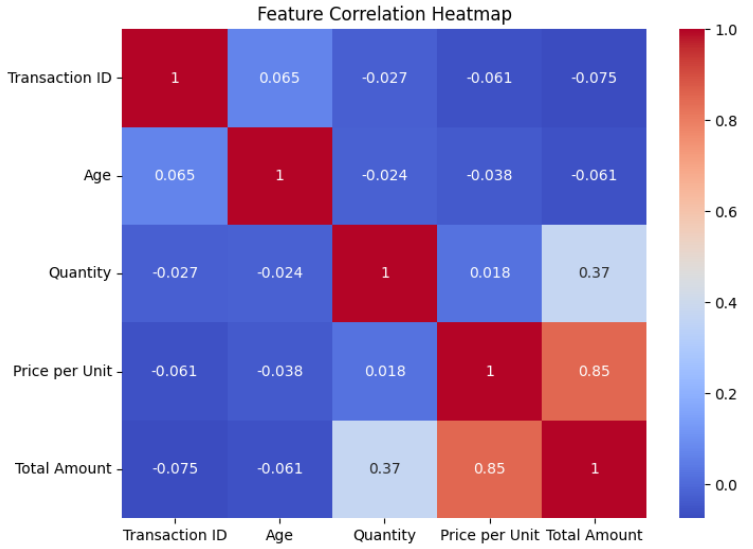
```
import seaborn as sns
import matplotlib.pyplot as plt

sns.pairplot(df)
plt.show()
```



Correlation Heatmap

[53]
✓ 0s

```
import seaborn as sns
import matplotlib.pyplot as plt

numeric_df = df.select_dtypes(include=['int64', 'float64'])

plt.figure(figsize=(8,6))
sns.heatmap(numeric_df.corr(), annot=True, cmap="coolwarm")
plt.title("Feature Correlation Heatmap")
plt.show()
```



PREPROCESSING

[54]
```
df = df.drop(['CustomerID', 'Date'], axis=1, errors='ignore')
```

[55]
```
df = df.fillna(df.mean(numeric_only=True))
```

[56]
```
from sklearn.preprocessing import LabelEncoder

le = LabelEncoder()

for col in df.select_dtypes(include=['object']).columns:
    df[col] = le.fit_transform(df[col])
```

select features for clustering

[57]
```
X = df[['Quantity', 'Price per Unit', 'Total Amount']]
```

[58]
```
from sklearn.preprocessing import StandardScaler

scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
```

ELBOW METHOD (WITH VISUAL)

[59]

```python
from sklearn.cluster import KMeans
import matplotlib.pyplot as plt

wcss = []

for k in range(1, 11):
    kmeans = KMeans(n_clusters=k, random_state=42)
    kmeans.fit(X_scaled)
    wcss.append(kmeans.inertia_)

plt.plot(range(1,11), wcss, marker='o')
plt.xlabel("Number of Clusters")
plt.ylabel("WCSS")
plt.title("Elbow Method")
plt.show()
```

APPLY K-MEANS AND LABEL DATA

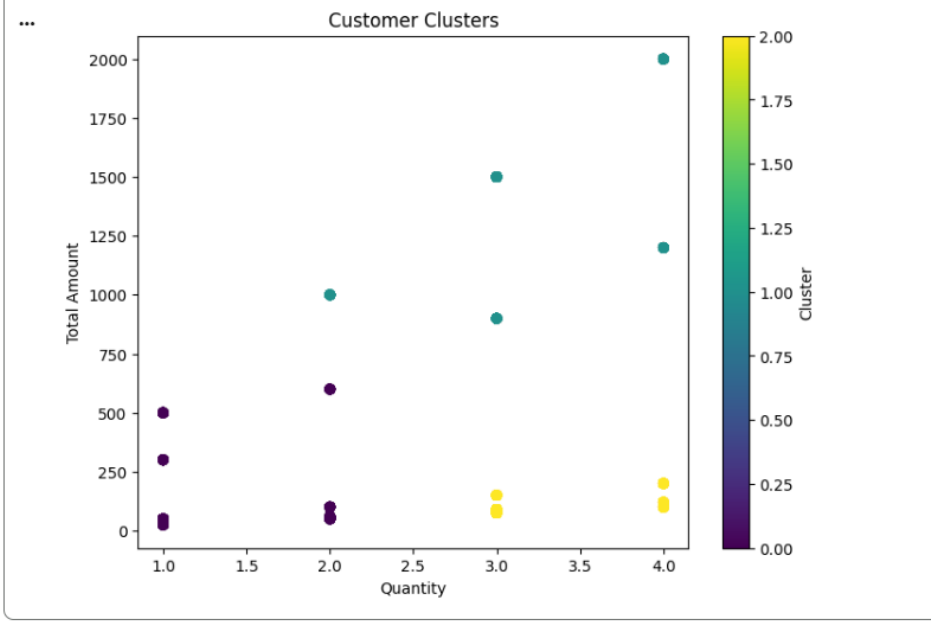
[60]

```python
k = 3
kmeans = KMeans(n_clusters=k, random_state=42)

df['Cluster'] = kmeans.fit_predict(X_scaled)
print(df.head())
```

```
   Transaction ID  Customer ID  Gender  Age  Product Category  Quantity  \
0               1            0       1   34                 0         3
1               2            1       0   26                 1         2
2               3            2       1   50                 2         1
3               4            3       1   37                 1         1
4               5            4       1   30                 0         2

   Price per Unit  Total Amount  Cluster
0              50           150        2
1             500          1000        1
2              30            30        0
3             500           500        0
4              50           100        0
```

[61]

```python
plt.figure(figsize=(8,6))
plt.scatter(df['Quantity'], df['Total Amount'], c=df['Cluster'])
plt.xlabel("Quantity")
plt.ylabel("Total Amount")
plt.title("Customer Clusters")
plt.colorbar(label="Cluster")
plt.show()
```



[62]

```
plt.figure(figsize=(10,6))
sns.boxplot(x=df['Cluster'], y=df['Total Amount'])
plt.title("Total Amount Distribution Across Clusters")
plt.show()
```

[64]

```
cluster_summary = df.groupby('Cluster').mean().round(2)
print(cluster_summary)
```

```
         Transaction ID  Customer ID  Gender    Age  Product Category  \
Cluster
0                505.88       505.78    0.51  41.90              1.07
1                468.84       468.74    0.48  40.14              1.02
2                521.10       517.89    0.46  41.75              0.99

         Quantity  Price per Unit  Total Amount
Cluster
0            1.43          136.09        171.94
1            3.20          412.12       1297.73
2            3.55           35.50        126.45
```

[65]

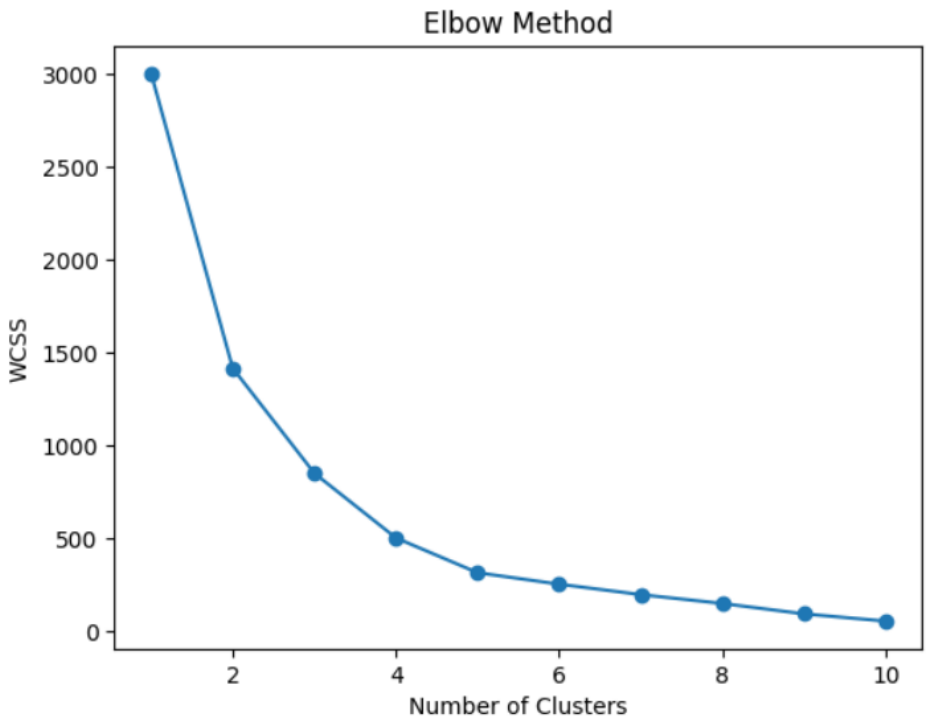
```
X = df[['Age', 'Gender', 'Quantity']]
y = df['Cluster']
```

[66]

```
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

model = RandomForestClassifier(random_state=42)
model.fit(X_train, y_train)

y_pred = model.predict(X_test)

print("Accuracy:", accuracy_score(y_test, y_pred))
```

```
Accuracy: 0.64
```

Output




```
new_customer = [[30, 1, 7]]   # Age, Gender, Quantity
predicted_cluster = model.predict(new_customer)

print("Predicted Cluster:", predicted_cluster)
```

```
Predicted Cluster: [2]
/usr/local/lib/python3.12/dist-packages/sklearn/utils/validation.py:2739: UserWarning: X does not have valid feature names, but RandomForestClassifier was fitted with feature names
  warnings.warn(
```

```
new_customer = [[30, 0, 7]]   # Age, Gender, Quantity
predicted_cluster = model.predict(new_customer)

print("Predicted Cluster:", predicted_cluster)
```

```
Predicted Cluster: [1]
/usr/local/lib/python3.12/dist-packages/sklearn/utils/validation.py:2739: UserWarning: X does not have valid feature names, but RandomForestClassifier was fitted with feature names
  warnings.warn(
```

```
new_customer = [[20, 1, 2]]   # Age, Gender, Quantity
predicted_cluster = model.predict(new_customer)

print("Predicted Cluster:", predicted_cluster)
```

```
Predicted Cluster: [0]
/usr/local/lib/python3.12/dist-packages/sklearn/utils/validation.py:2739: UserWarning: X does not have valid feature names, but RandomForestClassifier was fitted with feature names
  warnings.warn(
```

```
new_customer = [[60, 0, 10]]   # Age, Gender, Quantity
predicted_cluster = model.predict(new_customer)

print("Predicted Cluster:", predicted_cluster)
```

```
Predicted Cluster: [1]
/usr/local/lib/python3.12/dist-packages/sklearn/utils/validation.py:2739: UserWarning: X does not have valid feature names, but RandomForestClassifier was fitted with feature names
  warnings.warn(
```

Conclusion

The experiment demonstrated the integration of **unsupervised and supervised learning**:

- K-Means successfully segmented customers based on purchasing patterns.
- The Elbow Method helped identify the optimal number of clusters.
- Cluster labels were used for classification with Random Forest, achieving good prediction performance.
- This workflow highlights how clustering can generate labels for unlabeled data, enabling predictive modeling on future data.